CHAPTER 2:
OPERATING SYSTEM AND APPLICATION SOFTWARE

INTENDED LEARNING OUTCOMES(ILOS)
AT THE END OF THIS LESSON, THE STUDENT IS EXPECTED TO:

- UNDERSTAND HOW OPERATING SYSTEM WORKS;
- BE ABLE TO ENUMERATE THE ROLES OF OPERATING SYSTEM;
- EXPLAIN AND JUSTIFY THE PURPOSE OF AN OPERATING SYSTEM ;
- DIFFERENTIATE THE TYPES OF OPERATING SYSTEM INTERFACES;
- UNDERSTAND SOME BASIC FUNCTIONS OF OPERATING SYSTEM;
- HAVE A GENERAL OVERVIEW ON HOW PLUG AND PLAY TECHNOLOGY WORKS;
- UNDERSTAND THE USE OF APPLICATIONS PROGRAMMING INTERFACE;
- ENUMERATE SOME OF THE CAPABILITIES OF AN OPERATING SYSTEM;

INTENDED LEARNING OUTCOMES(ILOS)
AT THE END OF THIS LESSON, THE STUDENT IS EXPECTED TO:

- DESCRIBE THE DESKTOP OPERATING SYSTEM;
- ENUMARATE THE CHARACTERISTICS OF A DESKTOP OEPRATING SYSTEM;
- EXPLAIN HOW THE NETWORK OPERATING SYSTEM WORKS;
- LIST ALL THE COMMON TASK FOR NETWORK OPERATING SYSTEM;
- ENUMERATE THE COMMONLY USED DESKTOP OPERATING SYSTEMS;
- DIFFERENTIATES THE SIMILARITIES OF OPERATING SYSTEMS;
- LIST ALL THE ATTRIBUTES OF NETWORK OPERATING SYSTEM;
- ENUMERATE THE MOST COMMON NETWORK OPERATING SYSTEMS;
- DETERMINE OPERATING SYSTEM BASED ON CUSTOMER NEEDS;

INTENDED LEARNING OUTCOMES(ILOS)
AT THE END OF THIS LESSON, THE STUDENT IS EXPECTED TO:

- IDENTIFY WHAT DOES A CUSTOMER REQUIRE;
- ENUMERATE THE CUSTOMER NEEDS;
- IDENTIFY MINIMUM HARDWARE REQUIREMENTS ;
- UNDERSTAND THE HARDWARE COMPATIBILITY LIST;

OPERATING SYSTEM

- The operating system is the most important program that runs on a computer.
- Computer operating system perform basic tasks, such as recognizing input from the keyboard, sending output to the display, keeping files and directories on the storage drives, and controlling peripheral devices, such as printers.

Four Main Roles of operating system

- Manage hardware
- Run applications
- Provide a user interface
- Manage files

Purpose of an Operating System

- The operating system boots the computer and sets up the file system.

2 types of operating system interfaces:

- Command Line Interface (CLI): The user types commands at a prompt.
- Graphical User Interface (GUI): The user interacts with menus and icons.

Operating System Functions:

- File and folder management - The operating system creates a file structure on the hard disk to drive to allow data to be stored
- Application management - The operating system manage all programs to ensure that the correct resources are allocated to the applications.
- Control hardware access - The operating system handles the interaction between the applications and hardware.
- User interface - The operating system enable the user to interact with software and hardware.

Plug and Play Technology

- The operating system is designed to automatically discover new plug-and-play (PnP) compatible hardware, and then configure the device, OS, and registry.

Applications Programming Interface:

- The API allows programmers to create software applications consistent with the operating system.
- Open Graphics Library (OpenGL): Cross-platform standard specification for multimedia graphics.
- DirectX: Collection of API's related to multimedia tasks for Microsoft Windows.

Additional Capabilities of an Operating System

Almost all modern operating system can support more than one user, task, or CPU.

- Multi-user - A type of operating system that allows multiple users to access the operating system at the same time.
- Multitasking - A method a computer uses to share resources between multiple tasks, or process.
- Multiprocessing - A type of computer that can support more than one physical processor or the ability to schedule tasks between multiple CPU's.
- Multithreading - A method used by a CPU to process multiple threads, or tasks within a program, at the same time.

Desktop Operating System

A desktop operating system is intended for use in a small office/home office (SOHO) with a limited number of users. A network operating system (NOS) is designed for a corporate environment serving multiple users with a wide range of needs.

Characteristics of computer system:

- Designed to support a single user
- Designed to run single-user applications
- Designed to share files and folders on a small network
- Designed to share peripherals on a small network

Network Operating System

Characteristics of NOS:

- Designed to support multiple users
- Designed to run multi-user applications
- Designed to be robust and redundant
- Designed to be used on a network
- Designed with increased security compared to desktop operating systems

Common tasks for a NOS:

- File Transfer Protocol (FTP)
- Simple Mall Transfer Protocol (SMTP)
- Hypertext Transfer Protocol (HTTP)
- Lightweight Directory Access Protocol (LDAP)
- Active Directory
- Network operating systems are designed to function well in a client/server environment
- A server is a robust computer that is used to service the needs of multiple clients within a network.

Commonly used desktop operating systems fall into three group:

- **Windows**

Windows dominates the personal computer world, offering a graphical user interface (GUI), virtual memory management, multitasking, and support for many peripheral devices

- **Mac**

Mac OS features a graphical user interface (GUI) that utilizes windows, icons, and all applications that run on a Macintosh computer have a similar user interface.

- **Linux**

Linux is a freely distributed open source operating system that runs on a number of hardware platforms.

Operating systems have several similarities:

- GUI Interface
- Standard “look and feel” characteristics
- Multithreading capabilities
- Multitasking capabilities
- Ability to function with most hardware

Differences between desktop operating systems are typically related to availability and how much can be accomplished using the GUI:

- Windows and MAC OS users can perform the majority of tasks through the GUI.
- Linux and UNIX users most understand the use of CLI to perform some tasks.

The code for an operating systems will be either open source or proprietary:

- Open source applications can be read and modified. Programmers openly share code with other programmers. Linux distributions are open source.
- Proprietary application cannot be read or modified. Proprietary software agreements restrict the use of the software, identifying where and hen the software me be used.

Attributes of NOS

- A NOS has much of the same functionality as a desktop operating system.
- Determining the number of users that a server can support depends on factors such as the hardware specifications, the network operating system, and physical demands on the system.
- A NOS will normally remain stable when the number of users is high, but the process may become slow.

The most common network operating systems include:

- Novell Netware
- Microsoft Windows Server
- Linux
- UNIX

Determine operating system based on costumer needs

- Select the proper operating system to meet the needs of your customer, gain as much information about the costumer's daily activities as possible.
- Once you have a thorough understanding of what the customer needs, you can successfully select appropriate software and hardware to satisfy existing and future requirements.

What Does Your Customer Require?

To identify applications that customers will use and ensure compatibility

- When selecting hardware and software, the needs of your customer should come first. Remain neutral and listen to the needs of your customer before deciding on any computer hardware or software solution.

Explore customer needs by asking the following types of questions:

- What general office applications, such as word processing, spreadsheets, or presentation software, does your customer require?
- What graphics, such as Photoshop or illustrator, does your customer require?
- What animation, such as Flash, does your customer require?
- What business applications, such as accounting, contact management, sales tracking or database, does your customer require?

Once you have determined the types of software your customer requires, you should determine which available products will meet these needs. Before recommending applications, you should review the existing operating systems and software applications that your customer is using. The following are some of the factors to consider:

- Is compatibility an issue?
- Which operating system is required for current applications?
- Will the applications work with the existing hardware?
- Do any files have to be transferred between systems?
- Is there an issue with the format of the file systems involved?
- Are the applications standalone or networked?

As an example, your customer may already have invested in Windows-based business applications. In this case, there would be no need to recommend any other operating system, such as Linux. As a technician, you should recommend a solution that will benefit your customer and be cost-effective. If the customer has not invested in any one solution, then you will have more choices for the recommended technology.

Identify Minimum Hardware Requirements

- Operating systems and applications have minimum hardware requirements that must be met for the computer to be functional. In some cases, the application requirements may exceed the requirements of the operating system. For the application to function properly, it will be necessary to satisfy any additional requirement.

- Meeting only the minimum requirements may not be beneficial to your customer in the long term. Your customer may need to upgrade or purchase additional hardware.

Increasing the following are some common upgrades to the minimum requirements:

- RAM capacity
- Hard drive size
- Processor speed
- Video card memory and speed

- Identify the equipment that your customer has in place when you are analyzing hardware requirements. If upgrades are required to meet minimum requirements, conduct a cost analysis to determine the best course of action. In some cases, it may be necessary for the customer to purchase new equipment. In other cases, it may be cost-effective to upgrade the existing equipment.

Hardware Compatibility List (HCL) Most of operating systems have a hardware compatibility list (HCL) that can be found on the manufacturer's website. These lists provide a detailed inventory of hardware that has been tested and is known to work with the operating system. If the hardware already in place is not on the list, the hardware may require upgrading. If a hardware component is not on the list, there may be problems once installed.